WORKED SOLUTION

STEP I 2016, Question 8

Pure Mathematics • Generating functions

STEP I 2016, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q8

This question has a full guided treatment in *Beyond STEP*, Section 9.4, *Sequences as Functions*: https://beyondstep.keithmcnulty.org/ch09/generating-functions.html

The key idea

A generating function packages a whole sequence into a single function, and operations on the sequence become algebra on the function. In part (i) the binomial series for negative powers of $1-x$ produces the coefficients we want, and we build n^2 out of sequences we already know. In part (ii) we turn a recurrence into an equation for f by multiplying by x^n and summing: shifting the index of a sequence is the same as multiplying its generating function by x.

Solution

Part (i)

By the binomial series with index -2,

$$(1-x)^{-2} = \sum_{n\geq 0} \binom{-2}{n}(-x)^n = 1 + 2x + 3x^2 + 4x^3 + \cdots = \sum_{n\geq 0}(n+1)x^n,$$

since $\binom{-2}{n}(-1)^n = \frac{2\cdot 3\cdots(n+1)}{n!} = n+1$. Multiplying by x,

$$x(1-x)^{-2} = \sum_{n\geq 0}(n+1)x^{n+1} = \sum_{n\geq 0} nx^n,$$

where re-indexing the sum lets us include $n=0$ harmlessly (its term is 0). So $u_n = n$ has generating function $x(1-x)^{-2}$, as required.

Similarly, with index -3,

$$\binom{-3}{n}(-1)^n = \frac{3\cdot 4\cdots(n+2)}{n!} = \frac{(n+1)(n+2)}{2},$$

so

$$(1-x)^{-3} = \sum_{n\geq 0} \frac{(n+1)(n+2)}{2} x^n.$$

Multiplying by x and re-indexing as before,

$$x(1-x)^{-3} = \sum_{n\geq 0} \frac{n(n+1)}{2} x^n,$$

so the sequence is $u_n = \frac{1}{2}n(n+1)$: that is, $0, 1, 3, 6, 10, \ldots$, the triangular numbers.

Ask yourself How can we get n^2 from the two sequences we now know?

Generating functions are linear: if u_n and v_n have generating functions f and g, then $\alpha u_n + \beta v_n$ has generating function $\alpha \mathrm{f} + \beta \mathrm{g}$. Since

$$n^2 = 2 \cdot \tfrac{1}{2}n(n+1) - n,$$

the generating function for $u_n = n^2$ is

$$\frac{2x}{(1-x)^3} - \frac{x}{(1-x)^2} = \frac{2x - x(1-x)}{(1-x)^3} = \frac{x(1+x)}{(1-x)^3}.$$

Part (ii)(a)

For each $n \geq 1$ we have $u_n x^n = k u_{n-1} x^n$. Summing over all $n \geq 1$:

$$\sum_{n\geq 1} u_n x^n = kx \sum_{n\geq 1} u_{n-1} x^{n-1}$$
$$\mathrm{f}(x) - u_0 = kx\, \mathrm{f}(x),$$

because the left-hand sum is $\mathrm{f}(x)$ without its constant term, and the right-hand sum, with $m = n-1$, is $\sum_{m\geq 0} u_m x^m = \mathrm{f}(x)$. As $u_0 = a$, this is $\mathrm{f}(x) = a + kx\,\mathrm{f}(x)$, as required. Hence

$$\mathrm{f}(x) = \frac{a}{1-kx}.$$

(As a check, expanding gives $a + akx + ak^2x^2 + \cdots$, and indeed $u_n = ak^n$.)

Part (ii)(b)

Multiply $u_n = u_{n-1} + u_{n-2}$ by x^n and sum over $n \geq 2$ (the recurrence only holds from $n = 2$, so this is where we must start):

$$\sum_{n\geq 2} u_n x^n = x \sum_{n\geq 2} u_{n-1} x^{n-1} + x^2 \sum_{n\geq 2} u_{n-2} x^{n-2}.$$

Ask yourself Which terms of f are missing from each of these three sums?

The left-hand side lacks the terms $u_0 + u_1 x$; the first sum on the right runs over $u_1 x + u_2 x^2 + \cdots$ and so lacks only u_0; the second sum is all of $\mathrm{f}(x)$. Therefore

$$\mathrm{f}(x) - u_0 - u_1 x = x\big(\mathrm{f}(x) - u_0\big) + x^2\,\mathrm{f}(x).$$

With $u_0 = 0$ and $u_1 = 1$ this becomes $\mathrm{f}(x) - x = x\,\mathrm{f}(x) + x^2\,\mathrm{f}(x)$, so

$$\mathrm{f}(x) = \frac{x}{1 - x - x^2}.$$

This is the generating function of the Fibonacci numbers $0, 1, 1, 2, 3, 5, 8, \ldots$.

Checks

- Expanding $x(1+x)(1-x)^{-3} = (x + x^2)(1 + 3x + 6x^2 + 10x^3 + \cdots)$ gives $x + 4x^2 + 9x^3 + 16x^4 + \cdots$, the squares, as it should.
- Long division of x by $1 - x - x^2$ gives $x + x^2 + 2x^3 + 3x^4 + 5x^5 + \cdots$.
- At $x = 0.1$: the sum $\sum n^2 (0.1)^n$ and $0.11/0.9^3 = 0.150891\ldots$ agree, and $\sum u_n (0.1)^n$ for the Fibonacci numbers matches $0.1/0.89 = 10/89 = 0.112359\ldots$. These were confirmed by computer, along with the first 30 coefficients of every generating function in the question.

Common slips

- Getting the binomial coefficients for negative indices wrong: the signs from $\binom{-r}{n}$ and from $(-x)^n$ cancel, so every coefficient is positive.
- In (ii)(b), summing from $n = 0$ or $n = 1$, where the recurrence does not hold, or forgetting to subtract the initial terms u_0 and $u_1 x$.
- Writing the triangular numbers as $\frac{1}{2}n(n-1)$: check $u_1 = 1$.

Beyond STEP

Factorising $1 - x - x^2 = (1 - \phi x)(1 - \psi x)$, with $\phi, \psi = \frac{1}{2}(1 \pm \sqrt{5})$, and using partial fractions together with part (ii)(a) gives Binet's formula $u_n = (\phi^n - \psi^n)/\sqrt{5}$. The same method solves any linear recurrence with constant coefficients, and generating functions are a central tool in combinatorics and probability (where they become probability generating functions). Grimmett and Welsh, *Probability: An Introduction*, develops the probability side; Wilf's *generatingfunctionology* is a lively, freely available book on the combinatorial side.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.